Water-storage deficits and attribution limits across Thailand: a multi-source robustness assessment

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Abstract

Separating climate-related storage change from human influence is difficult when satellite gravity, hydrological models and local observations have different spatial support and incomplete process representations. We assess this problem across Thailand's 22 principal basins using GRACE-based terrestrial and groundwater storage anomalies and a directly applied naturalized WaterGAP reference, without a statistical storage reconstruction. A factorial sensitivity analysis crosses gravity processing, soil-moisture subtraction, coastal support and reporting scale. An additional WaterGAP forcing variant provides a baseline sensitivity test; two other no-water-management models provide structural diagnostics. Independent evidence uses fixed groundwater-station panels and scene-supported reservoir storage. Daily CHIRPS records for 2003–2023 support an exploratory rainfall-concentration analysis controlling annual rainfall, potential evaporation and time. Negative residual slopes persist in all tested product/component variants for 20 TWSA basins and 20 GWSA basins. National total-storage residual Sen slopes across the two WaterGAP variants range from -8.93 to -8.91 mm/year on common 2002–2021 support. The balanced groundwater panel yields r=-0.00 across 8 usable years, whereas the adjusted national reservoir association is r=0.44 across 73 months. These tests distinguish evidence for storage deficits from evidence for a particular human cause. The combined residual cannot be interpreted as measured abstraction.

1 Introduction

Thailand's monsoon-driven water system combines seasonal recharge, groundwater use, irrigated agriculture and regulated surface water. Satellite gravity provides regional total-water-storage observations, while hydrological simulations provide physically structured counterfactuals. Their difference is useful for diagnosing unresolved change, but a naturalized simulation is not an observation of a world without human influence. Previous Chao Phraya studies already examined storage components, extremes and groundwater reliability [1,2]. Climate-reconstruction subtraction is also established in the literature [3]. Accordingly, this study does not claim novelty from basin trend maps or subtraction alone.

The contribution tested here is the robustness and identifiability of Thailand's water-storage deficits across processing and baseline choices, with independent component evidence. Recent work identifies scale-dependent model–observation inconsistencies [4], the need for GRACE-compatible compartment support

[5], and the value of comparing groundwater budgets with satellite estimates [6]. Rainfall concentration offers an additional process hypothesis: storage can respond to the distribution of daily rainfall as well as its total [7]. We ask which negative residuals persist, which receive observational support, and which remain unresolved.

2 Data and methodology

GRACE-based TWSA and component-derived GWSA are expressed in equivalent water thickness, referenced to 2004–2009. GWSA is TWSA minus soil moisture, surface water, canopy water and snow. Monthly comparisons retain missing months and use common grid/date support within each experiment. No observational mission gaps are filled. The primary naturalized WaterGAP v2.2e baseline is used directly; no residence-time model, learned emulator or climate reconstruction enters the subtraction [8].

Let O denote observed/derived storage, C naturalized WaterGAP, and A historical WaterGAP. The accounting is R=O-C=(A-C)+(O-A)=H+E. H is a model-estimated human perturbation and E the remaining model–observation discrepancy. Neither the identity nor a correlation with another variable establishes causal identification. GLDAS comparisons are restricted to matching soil, canopy and snow compartments, because the available Noah land-storage sum is not full TWSA.

The processing experiment crosses CSR and COST-G/G3P gravity with Noah and G3P root-zone soil estimates for GWSA, holding other subtraction components fixed. TWSA has two gravity variants. The same valid grid/month mask and reference months are used across variants. A one-cell interior-mask erosion is a coastal sensitivity test, not a validated leakage correction. Results are also aggregated to five explicitly listed geographic reporting groups and Thailand; grouping does not create independent satellite observations. Seasonal Sen slopes compare observations from the same calendar month. Month-adjusted OLS slopes have two-year calendar-block wild intervals, conditional on fixed data/support. Product sensitivity uses 199 replicates; alternative-baseline comparisons use 499. These conditional intervals do not represent the complete product/model uncertainty budget.

The baseline comparison uses WaterGAP/GSWP3–ERA5 and three public ISIMIP3a obsclim histsoc nowatermgt TWS simulations: WaterGAP, LPJmL5-7-10-fire and MIROC-INTEG-LAND driven by 20CRV3–ERA5. All use exact common 2002–2021 support and a common anomaly reference. The shared forcing helps compare the additional model simulations, while the primary WaterGAP differs in forcing and experiment setup. Historical land use, CO2 or other forcings may remain in nowatermgt runs; thus these are sensitivity references, not four identical pristine-world counterfactuals. MIROC represents groundwater processes, but the downloaded TWS field lists soil, canopy, snow, river and reservoir storage without a separately identified groundwater term. LPJmL also requires explicit version-specific groundwater-accounting verification. Until compartment equivalence is established, both are retained as structural diagnostics and excluded from full-TWS baseline evidence flags. Thus a fully compatible, structurally independent naturalized-model ensemble is not claimed. No GRACE-assimilating product is treated as independent validation.

Groundwater observations are oriented so increasing head index means shallower water. Co-located wells are aggregated by coordinate group. Changing-network and strict ten-year-complete station-group panels are compared over 2013–2022; annual storage requires at least eight observed months. Station-bootstrap

intervals are conditional on the observed years. Reservoir tests use fixed pre-2004 reservoirs and scene-supported Level 1 storage [9]. Season/trend removal uses calendar-month indicators and linear time on paired dates. Year-cluster bootstrap intervals and leave-one-year-out influence ranges test temporal robustness. Reservoir changes include climate and operations and cannot directly validate groundwater or human-only storage.

Daily CHIRPS v2 is retrieved from the publisher at native 0.05-degree pixel centres for every calendar day in 2003–2023. Cosine-latitude weights aggregate cells by basin. The primary concentration metric is the basin area mean of local daily-rainfall Gini coefficients, computed per grid cell before aggregation. The Gini of basin-mean daily rainfall and local consecutive dry days (rainfall below 1 mm/day) are sensitivities. Complete annual coverage is required. Annual storage is deseasonalized with the 2004–2009 monthly reference and averaged only for years with at least eight paired months. Ordinary least squares controls annual rainfall, native potential ET and linear time. Coefficients are expressed per metric standard deviation; two-year wild bootstrap intervals and null tests use 999 draws. Leave-one-year-out explanatory comparisons standardize predictors within each training fold. These are diagnostic regressions, not a reconstructed climate baseline or an operational forecast. FDR adjustment includes every fixed basin test: 22 basins, two storage variables, three response components and three rainfall metrics; national tests form a separate family. The short annual record, derived daily rainfall and spatial dependence limit inference.

3 Results

Negative residual Sen slopes occur across every tested product/component variant in 20 of 22 TWSA reporting basins and 20 of 22 GWSA reporting basins. This is robustness within the tested choices, not a probability of human causation. The spatial range across variants and the interior-mask sensitivity identify where apparent magnitude is unstable. Regional aggregation provides a check on small-basin interpretation; it cannot validate a local hotspot.

On common support, the two WaterGAP variants give national residual Sen slopes spanning -8.93 to -8.91 mm/year. Baseline disagreements combine structure, forcing and scenario definitions, so the spread is reported as a sensitivity range rather than an independent ensemble confidence interval.

Independent evidence is mixed. The strict balanced national groundwater panel has 189 coordinate groups but only 8 usable storage years; the GWSA residual correlation is -0.004. The station-resampling interval is -0.12 to 0.13 and is conditional on those years. National scene-supported reservoir storage has an adjusted correlation of 0.439 with the TWSA residual (73 paired months), with a year-cluster interval of 0.19 to 0.65. The leave-one-year-out adjusted correlations range from 0.36 to 0.52. Reservoir evidence supports partial surface-water consistency; it does not resolve the origin of the full residual.

The rainfall-concentration experiment yields 0 basin tests with FDR q below 0.05 across the full specified family. The complete results retain positive and negative explanatory gains, and show observed, naturalized and residual responses separately. A relationship in the residual could indicate incomplete climate representation, other correlated processes or errors; its absence would not prove complete climate removal.

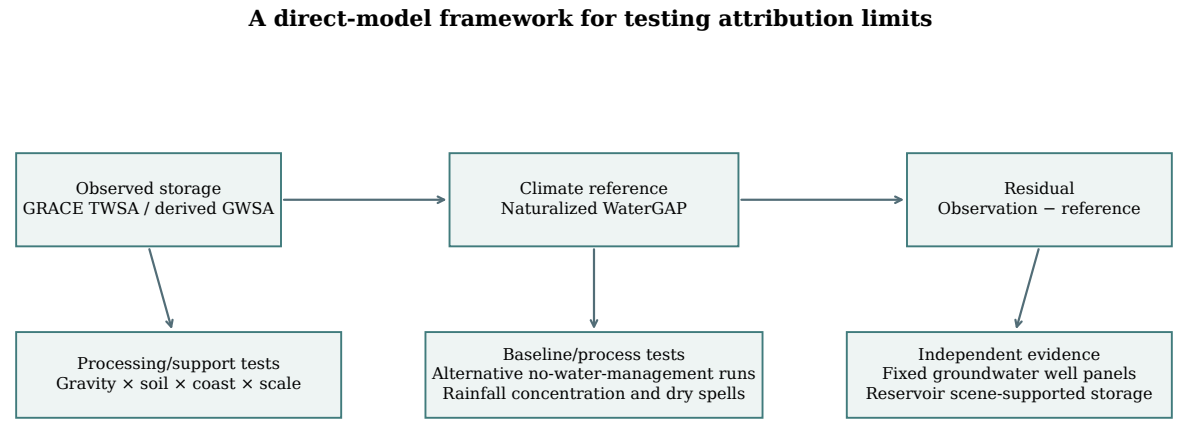


Figure 1: A direct-model framework for testing attribution limits. The analysis tests robustness and physical consistency; it does not identify all residual storage as human-caused. Climate reconstruction and residence-time fitting are excluded from this design.

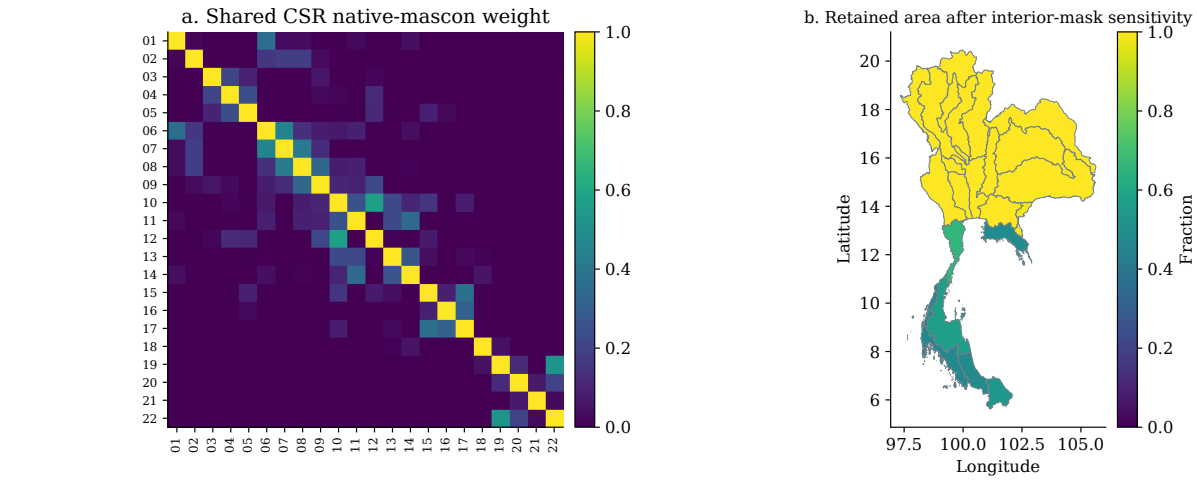


Figure 2: Why 22 reporting basins are not 22 independent satellite regions. Shared native weights describe overlapping sampling, not a full GRACE error covariance or independent effective sample size. Interior sensitivity removes a 0.25° grid-cell border next to provider ocean masks; it is a stress test, not a validated coastal correction.

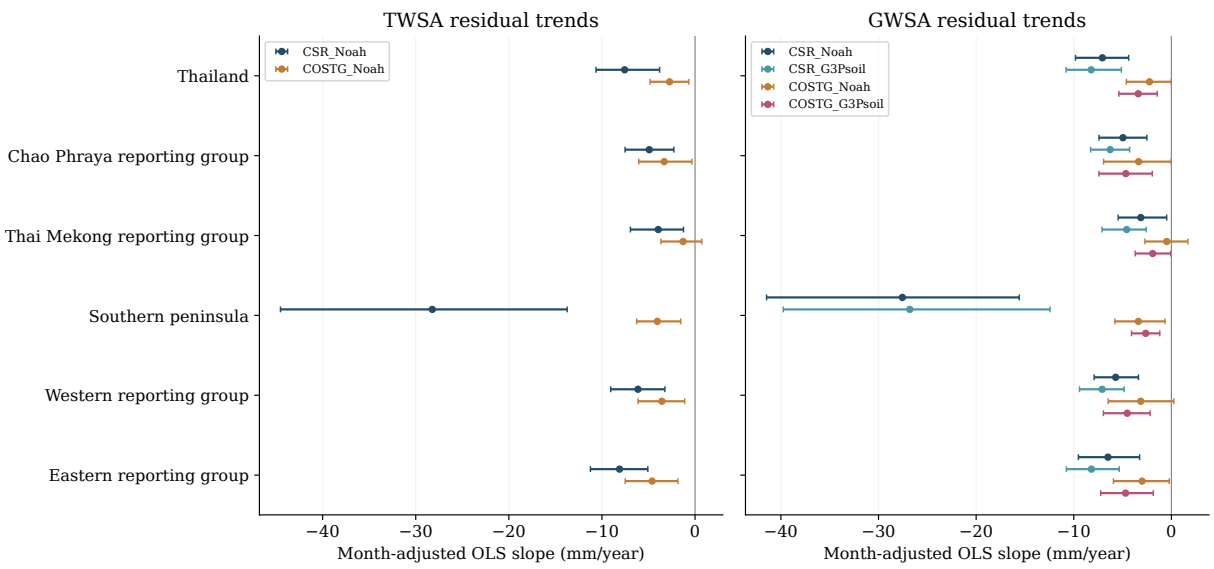


Figure 3: Do declines survive gravity and soil-component choices?. Intervals: conditional two-year calendar-block wild regression; they do not include all satellite/model uncertainty. Regional groups aggregate reporting basins, not new independent GRACE measurements; TWSA does not depend on soil choice.

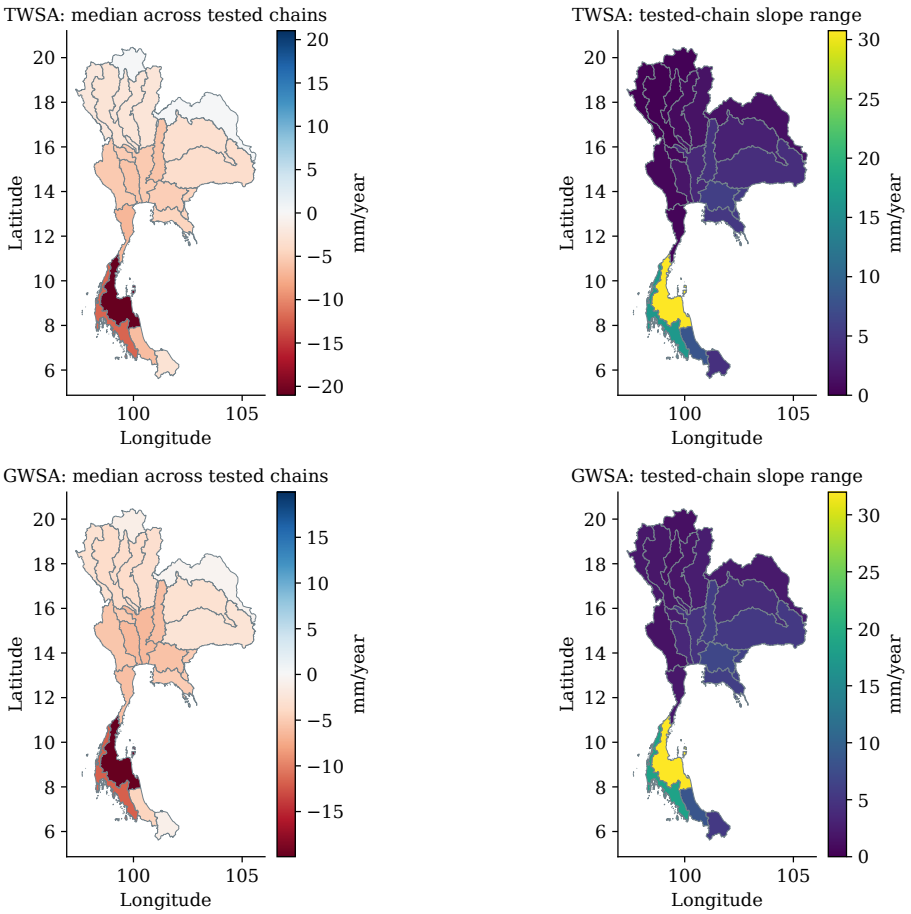


Figure 4: Residual trend magnitude and its processing sensitivity. Median/range across two gravity products and, for GWSA, two soil estimates. Other subtraction terms are held fixed. A narrow range within this tested set is not proof of small total uncertainty; basin-scale GRACE support remains coarse.

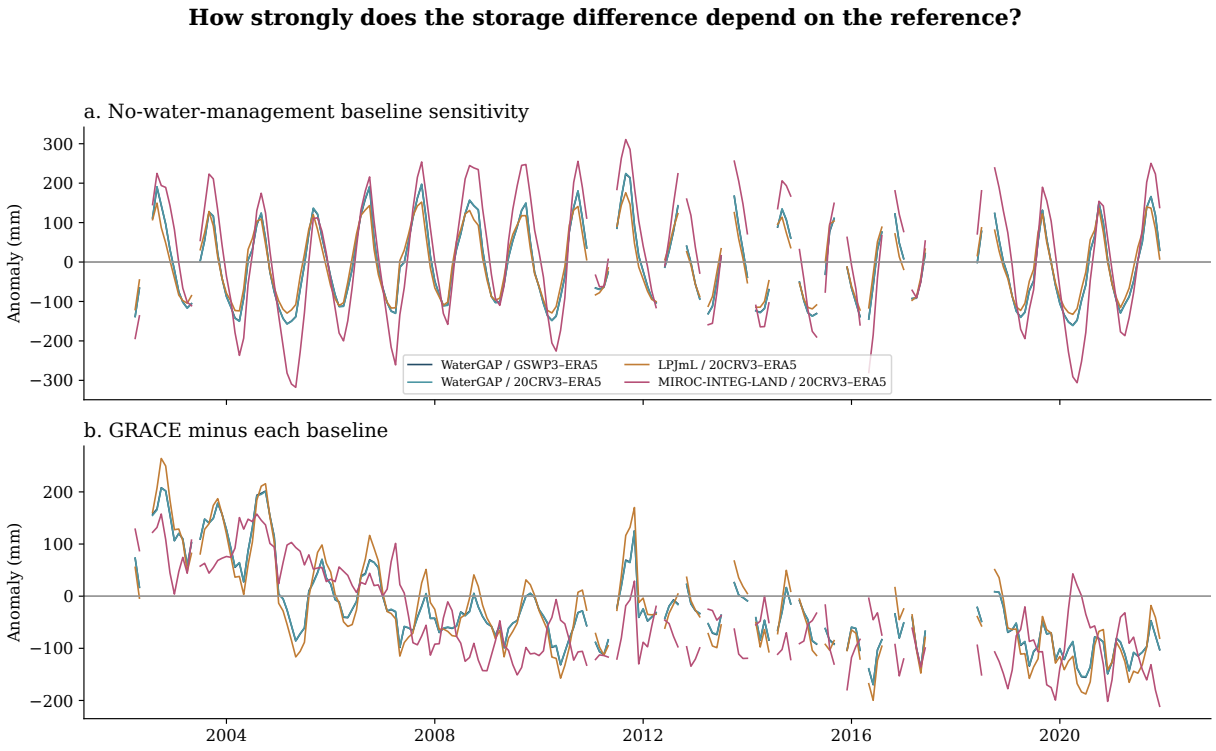


Figure 5: How strongly does the storage difference depend on the reference?. Exact common 2002–2021 support and 2004–2009 reference. LPJmL/MIROC are structural diagnostics, excluded from full-TWS evidence flags. Their groundwater accounting is not verified as equivalent to WaterGAP; nowatermgt can retain other historical human forcings.

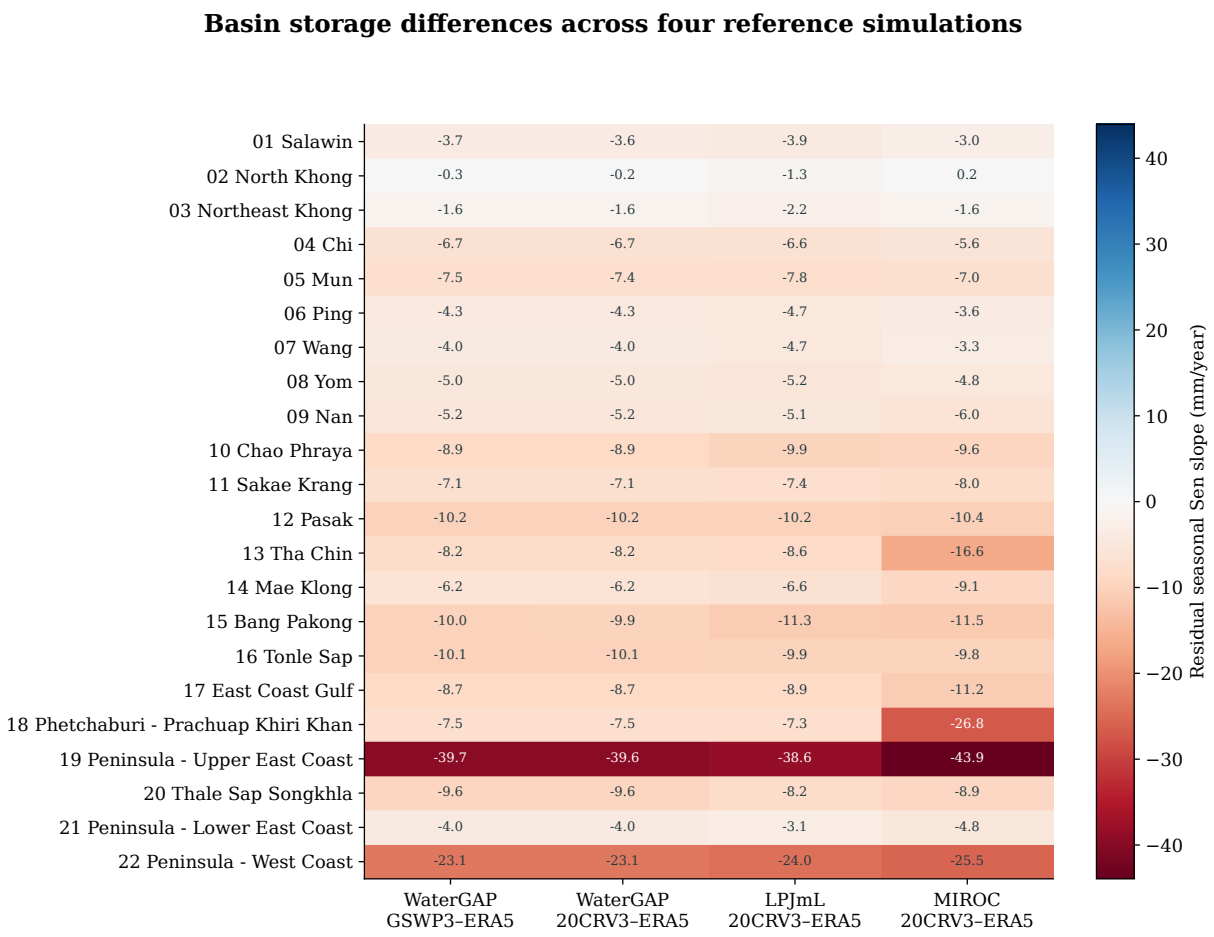


Figure 6: Basin storage differences across four reference simulations. TWSA differences, common 2002–2021 support. LPJmL/MIROC are structural diagnostics, not verified equivalent full-TWS baselines. Only the two WaterGAP variants enter baseline-robustness flags. Differences combine forcing, model structure and experiment definitions.

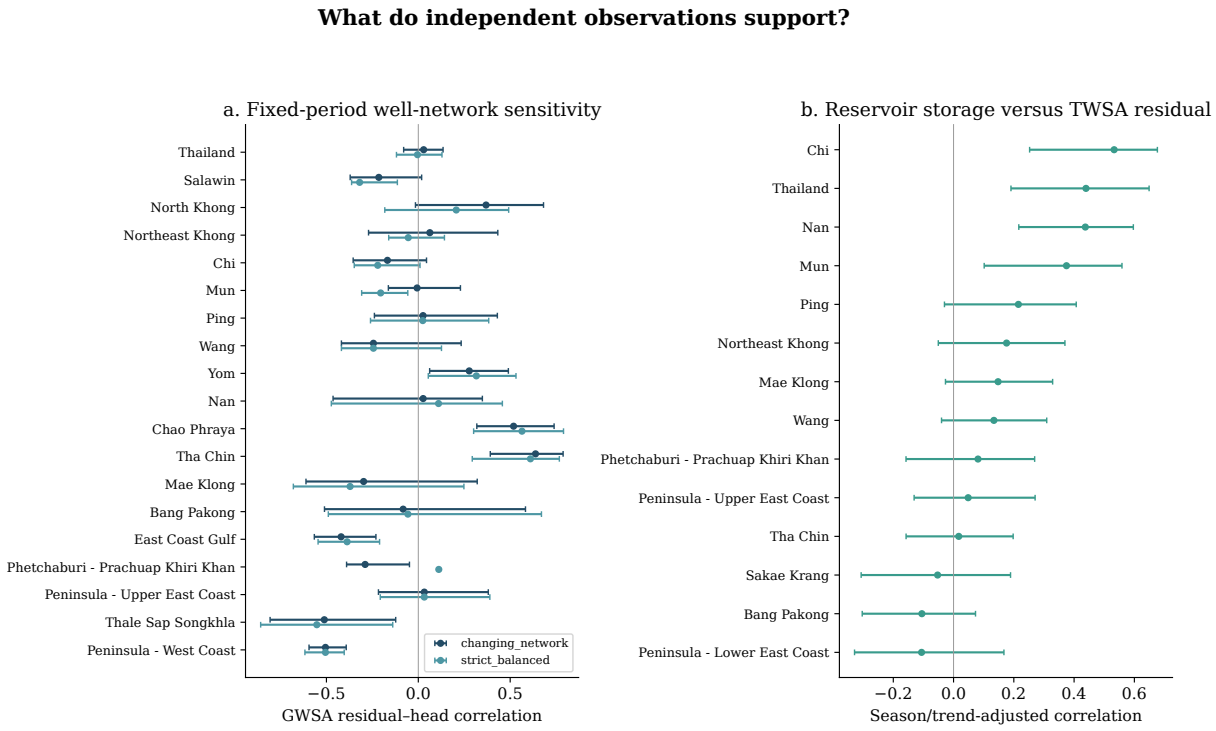


Figure 7: What do independent observations support?. Wells: 2013–2022, ≥8 paired years; station-bootstrap intervals conditional on observed years (changing/strict panels). Reservoirs: ≥24 scene-supported months; year-cluster intervals. Correlations are not causal attribution or storage-volume validation.

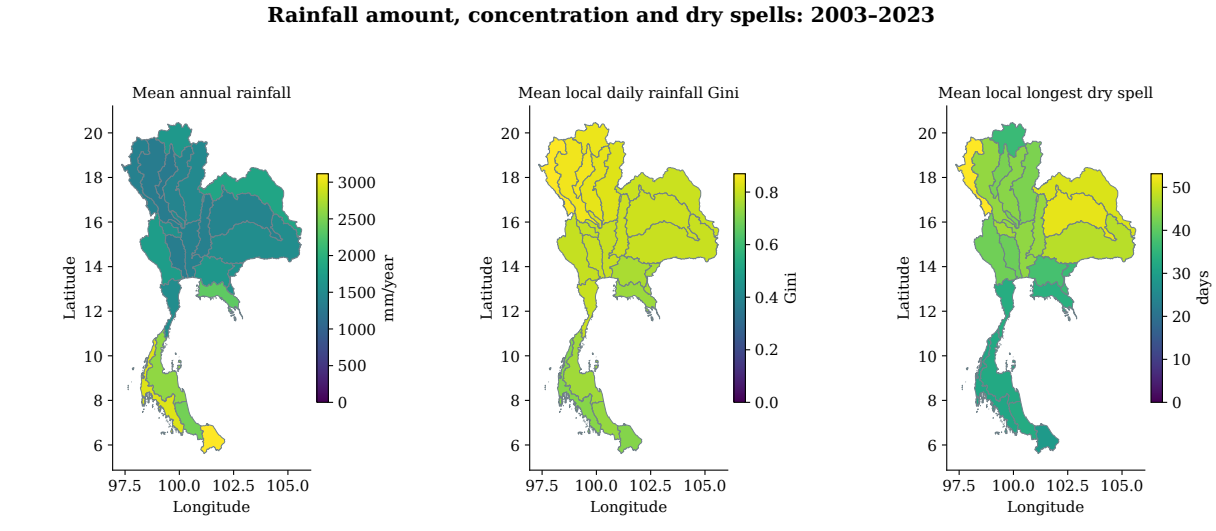


Figure 8: Rainfall amount, concentration and dry spells: 2003–2023. CHIRPS v2 daily at native 0.05° pixel centres, cosine-latitude area weights. Higher Gini means more concentrated daily rainfall. Local indices are calculated per grid cell before basin averaging; dry days have rainfall <1 mm. Daily CHIRPS is a derived rainfall product.

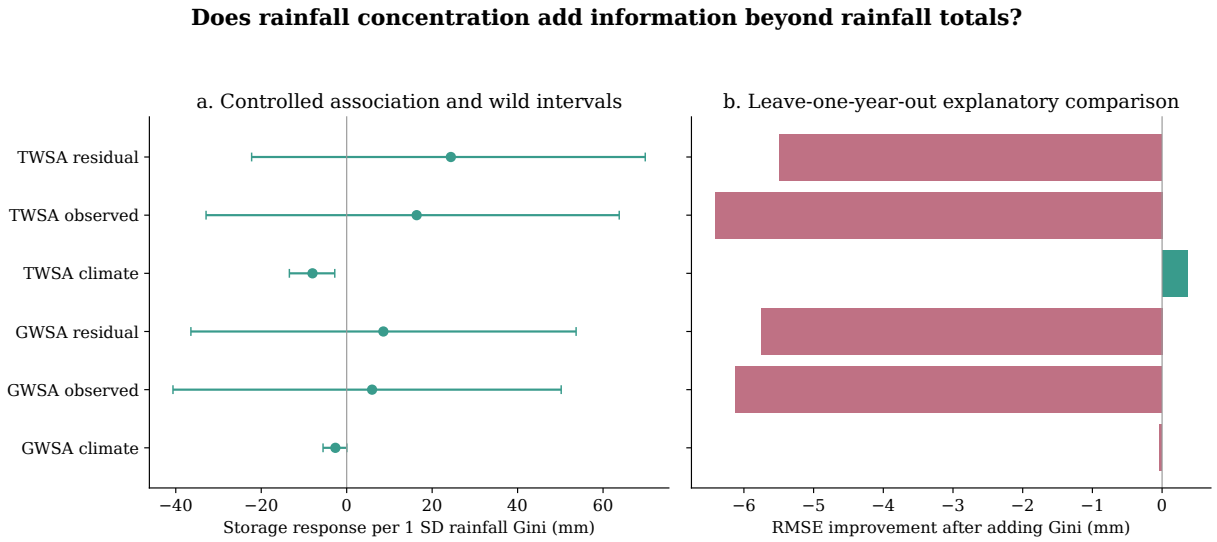


Figure 9: Does rainfall concentration add information beyond rainfall totals?. Controls: annual precipitation, annual potential ET and linear time; main predictor: area-mean local daily Gini. Two-year wild intervals; small annual samples. Positive RMSE gain means improvement. These are exploratory associations, not causal effects.

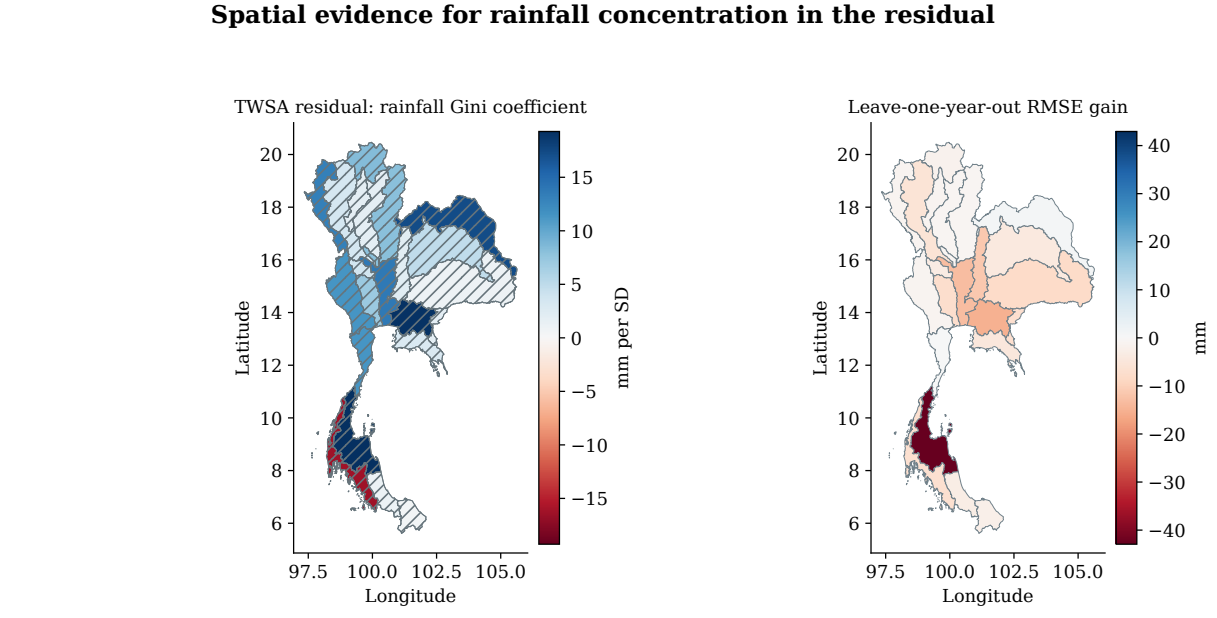


Figure 10: Spatial evidence for rainfall concentration in the residual. Hatching: $q \geq 0.05$ after adjustment across all fixed basin tests (both storage variables, three responses, three rain metrics). Basins share GRACE/climate signals. Improved explanatory fit alone does not establish a mechanism or anthropogenic influence.

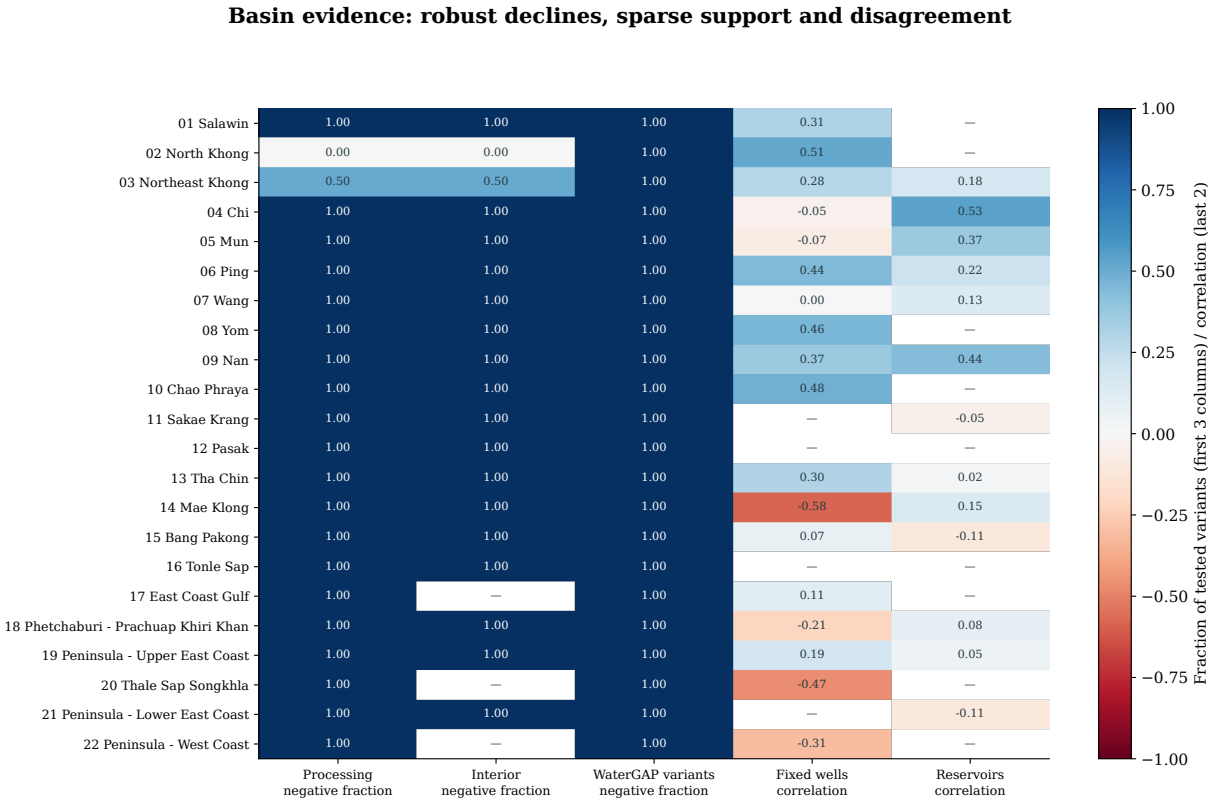


Figure 11: Basin evidence: robust declines, sparse support and disagreement. Fractions describe tested variants, not probabilities; baseline flags use only two WaterGAP variants, not the structural diagnostics. Correlations describe consistency, not causality. Missing evidence remains blank; no overall confidence score is assigned.

4 Discussion

The study's principal contribution is a transparent distinction between detecting a storage deficit and attributing its cause. A persistent GRACE–naturalized-model difference can survive selected processing changes while still lacking convincing independent evidence for pumping. Weak groundwater agreement is scientifically relevant and is not repaired by selecting favorable basins or lags. Groundwater head, aquifer storage and basin total storage represent different quantities. Specific yield or confined-aquifer storativity is required for defensible volume conversion, and an assumed uniform value is not introduced here.

GLDAS has a valuable but limited role. Agreement of matching soil/canopy/snow compartments is a cross-model check, not an independent observation of complete naturalized TWS. GLDAS also contributes to derived GWSA, creating dependence. Likewise, adding another model tests baseline sensitivity rather than identifying the true counterfactual. The rainfall analysis investigates a plausible omitted process but does not establish that concentration explains the residual. Its short annual sample and multiple tests demand restraint.

All 22 basins remain useful reporting units, but their nominal raster spacing is not GRACE's independent resolving scale. Shared native mascon weights and coastal sensitivity are reported explicitly. Future work should prioritize aquifer-stratified long monthly well records, metered abstractions, reservoir operations and independent groundwater budgets. A formally controlled management intervention could strengthen causal attribution if suitable data and assumptions exist. Those data are not claimed to exist in the present analysis.

The decision-support system therefore displays evidence and uncertainty alongside storage trends. It supports regional screening and monitoring priorities; it does not prescribe basin pumping limits from the residual. The novelty proposed here is conditional on the documented experiments and should not be described as a first-ever framework without a broader systematic review.

5 Conclusions

Direct use of naturalized WaterGAP permits a simple and auditable storage comparison without a climate reconstruction. The added experiments distinguish processing robustness, baseline sensitivity, exploratory climate-process evidence and independent component consistency. Negative residuals do not, by themselves, establish anthropogenic depletion. The national balanced groundwater comparison is weak, whereas reservoir evidence is stronger but partial. A defensible publication should emphasize these tested limits and the basin evidence matrix rather than report an unqualified human contribution percentage.

Data and code availability

All processed inputs, acquisition provenance, analysis scripts, figure sources and numerical reference tables are distributed with the reproducible research release. Baseline simulations remain publisher products, not models developed or recalibrated in this study. The analytical code rebuilds the new tables and figures without requiring the earlier reconstruction model.

Basin evidence table

Table 1: Basin residual sensitivity and independent evidence. Slope ranges are mm/year across tested processing choices; correlations are not causal validation. Codes correspond to named basin panels and the supplied CSV.

Code	TWS range	GWS range	Fixed GW r	Years	Reservoir r	Months
01	-2.5 to -2.4	-4.2 to -2.2	-0.32	8	—	0
02	0.1 to 0.3	-2.1 to -0.8	0.21	8	—	0
03	-0.5 to 0.9	-1.6 to 0.6	-0.05	8	0.18	179
04	-4.8 to -2.0	-4.8 to -0.8	-0.22	8	0.53	132
05	-5.7 to -1.6	-5.4 to -0.1	-0.20	8	0.37	135
06	-3.0 to -3.0	-4.6 to -2.6	0.02	8	0.22	106
07	-2.6 to -2.5	-4.1 to -2.8	-0.24	8	0.13	176
08	-3.1 to -2.2	-4.6 to -2.7	0.32	8	—	0
09	-3.5 to -1.9	-4.6 to -2.1	0.11	8	0.44	198
10	-7.2 to -3.7	-8.6 to -3.9	0.56	8	—	0
11	-5.6 to -5.0	-6.6 to -4.1	—	0	-0.05	192
12	-8.1 to -3.8	-8.5 to -2.8	—	0	—	0
13	-6.8 to -5.1	-8.2 to -4.6	0.61	8	0.02	189
14	-5.8 to -5.3	-6.2 to -4.7	-0.37	8	0.15	176
15	-8.0 to -2.3	-8.9 to -2.1	-0.06	8	-0.11	164
16	-7.4 to -2.0	-7.6 to -1.4	—	0	—	0
17	-7.2 to -2.1	-7.7 to -1.8	-0.39	8	—	0
18	-7.2 to -6.9	-7.1 to -5.1	0.11	8	0.08	170
19	-36.4 to -5.7	-36.8 to -4.8	0.03	8	0.05	171
20	-10.5 to -2.4	-8.7 to -0.2	-0.55	8	—	0
21	-5.2 to -1.0	-4.4 to 0.9	—	0	-0.11	201
22	-20.4 to -4.0	-21.9 to -3.5	-0.51	8	—	0

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